parent: therefore, we enter the loop at line #12, and perform a left rotation; after the first iteration of the loop and the left rotation, the “Beer” node still has higher priority than its new parent, “Butter”, as shown in the top half of figure 3.9. Therefore, we enter a second iteration of the loop, this time performing a right rotation, because node `X` is now a left child of `P'`.

Since now (bottom half of figure 3.9) no invariant is violated anymore, we can exit the loop; since the new node wasn’t bubbled up all the way to the root, the check at line #14 fails and we don’t need to do anything else.

What’s the running time for insert? Adding a new leaf requires `O(h)`, because we need to traverse the tree from its root to a leaf; then we can bubble up the new node at most to the root, and at each rotation we move the node one level up, so we can perform at most `h` rotations; each rotation, moreover, requires a constant number of pointers to be updated, so that bubbling up the new node, and the whole method, finally require `O(h)` steps.